CFD-Informed Linear-to-Nonlinear Instability Growth in High-Mach-Number Drop Aerobreakup

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Research Team in energy conversion & propulsion revolution

Motivation – Rain Erosion

- Hypersonic glide vehicles encounter atmospheric hydrometeors during terminal descent at 0–7 km
  - Micron cloud droplets to millimeter raindrops
- Cumulative droplet impacts roughen the surface → early boundary-layer transition → heating/drag/control-authority penalties
- These margins are already thin in air-breathing systems

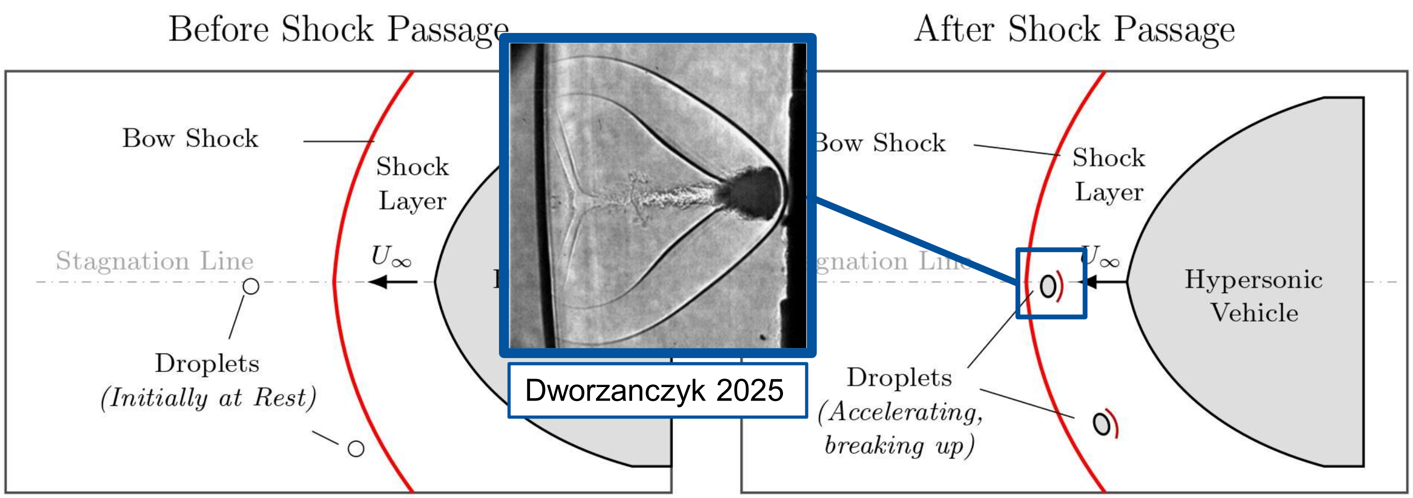
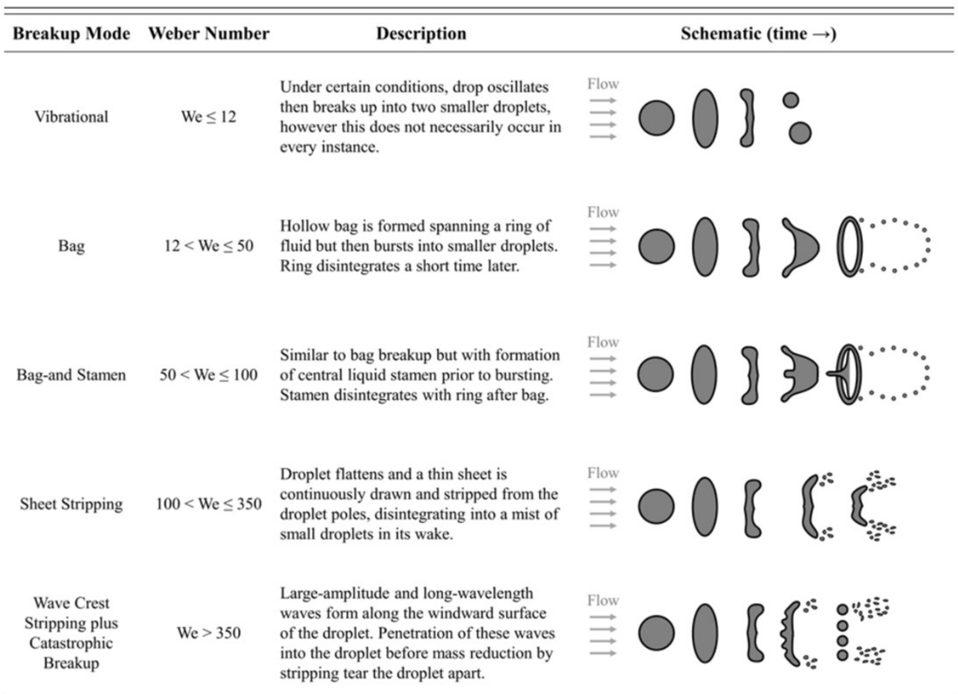


Figure: Schematic of atmospheric droplets interacting with the bow shock of a hypersonic vehicle before and after shock passage. Adapted from Aguilera 2023.

Introduction – Breakup Modes

- Droplet breakup is commonly organized by the aerodynamic Weber number: $We_A = \frac{\rho_g U_A^2 d_d}{\sigma}$



Breakup Mode	Weber Number	Description	Schematic (time →)
Vibrational	We ≤ 12	Under certain conditions, drop oscillates then breaks up into two smaller droplets, however this does not necessarily occur in every instance.	Flow
Bag	12 < We ≤ 50	Hollow bag is formed spanning a ring of fluid but then bursts into smaller droplets. Ring disintegrates a short time later.	Flow
Bag-and Stamen	50 < We ≤ 100	Similar to bag breakup but with formation of central liquid stamen prior to bursting. Stamen disintegrates with ring after bag.	Flow
Sheet Stripping	100 < We ≤ 350	Droplet flattens and a thin sheet is continuously drawn and stripped from the droplet poles, disintegrating into a mist of small droplets in its wake.	Flow
Wave Crest Stripping plus Catastrophic Breakup	We > 350	Large-amplitude and long-wavelength waves form along the windward surface of the droplet. Penetration of these waves into the droplet before mass reduction by stripping tear the droplet apart.	Flow

Figure: Breakup modes of water droplets observed experimentally (Aguilera 2023).

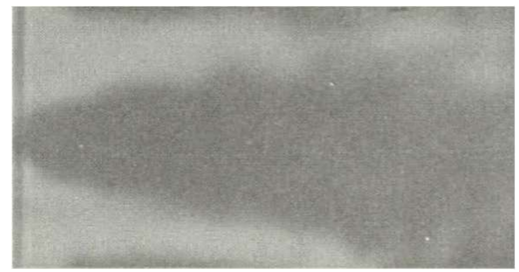


Figure: Droplet, Mach 3 (Reinecke-McKay 1969)

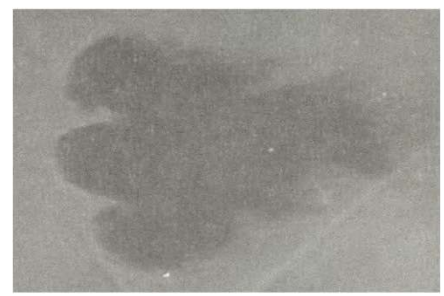


Figure: Droplet, Mach 11 (Reinecke-McKay 1969)

Literature: Linear-to-Nonlinear Instability Growth

- 2025 - Dworzanczyk et al. studied high-Mach drop aerobreakup using railgun experiments
- Applied early-time RT/KH linear stability analysis
  - Found linear theory was useful for locating instability mechanisms
  - Not useful for predicting breakup morphology

Linear growth rate: (based on Funada and Joseph 2001)

$$s = -i\kappa \frac{\rho_g u_g + \rho_d u_d}{\rho_g + \rho_d} - \frac{\kappa^2(\mu_g + \mu_d)}{\rho_g + \rho_d} \pm \left[\frac{\kappa^2 \rho_g \rho_d (u_g - u_d)^2}{(\rho_g + \rho_d)^2} - \frac{\kappa a_\perp (\rho_d - \rho_g)}{\rho_g + \rho_d} - \frac{\kappa^3 \sigma}{\rho_g + \rho_d} + \frac{\kappa^4 (\mu_g + \mu_d)^2}{(\rho_g + \rho_d)^2} + 2i\kappa^3 \frac{(\rho_g \mu_d - \rho_d \mu_g)(u_g - u_d)}{(\rho_g + \rho_d)^2} \right]^{1/2}$$

Nonlinear growth rate: (based on Pilch et al. 1987 and Davies and Taylor 1950)

$$\left.\frac{d\zeta}{dt}\right|_{nl} = F\sqrt{(1-\epsilon)a\lambda}$$

- Predicts finite-amplitude growth when wavelength is supplied

Unresolved issue: **wavelength is not predicted**

- Linear theory predicts $\lambda \sim 30\ \mu m$; experiments / simulations show structures $\sim 100\ \mu m\ to\ 1\ mm$

Problem Statement

Can CFD-informed inputs improve the prediction of the linear-to-nonlinear framework proposed by Dworzanczyk et al.?

Use **extracted base states** from **computational simulations** to update growth-rate maps, **assess other linear mechanisms**, and **identify simulation requirements** needed for future work.

Objectives:

1. Summarize the linear-to-nonlinear framework
2. Construct CFD-informed growth-rate maps
3. Assess wavelength-selection candidates systematically
4. Diagnose the structural limitation of the simulations

Methodology: Simulation Matrix

Parameter space: 1mm droplet, $M_s \in \{3, 5, 7\}$

Case	M_s	ρ_g/ρ_a	$U_A\ (m\ s^{-1})$	We_A
1	3	3.86	1,510	$\approx 1.5 \times 10^5$
2	5	5.00	2,895	$\approx 7.0 \times 10^5$
3	7	5.44	4,310	$\approx 1.7 \times 10^6$

Table: Shock-droplet simulation matrix. Post-shock values from Rankine-Hugoniot relations; We_A based on post-shock gas conditions and initial drop diameter.

- All We are well above those for the classical breakup mechanisms

Instability analysis:

- At each stored time step:
  - Extract ρ_g, u_g, u_d, and $a_\perp$, all $f = f(\phi, t)$
  - Evaluate $s_r(\kappa; \phi, t)$ and identify most-amplified wavenumber $\kappa^*(\phi, t)$

Methodology: Computational Setup

All simulations are performed in cylindrical coordinates (x, r) under an axisymmetric assumption.

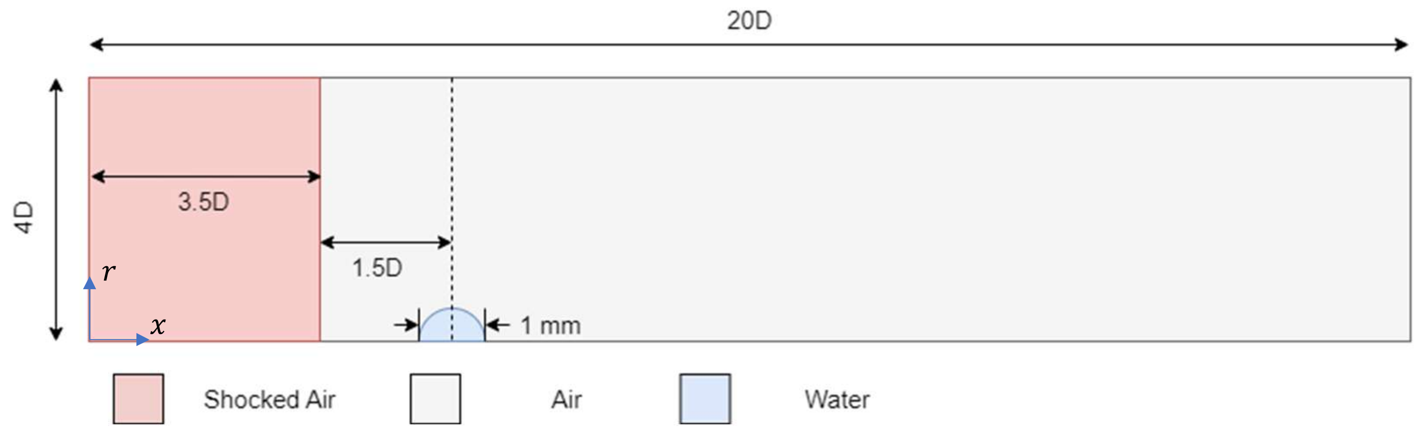


Figure: Computational domain, normalized by droplet diameter, *D*.

Boundary conditions:

- The bottom boundary is an axis of symmetry
- All other boundaries are non-reflecting boundaries

Initial conditions:

- Shocked air set by Rankine-Hugoniot jump conditions into quiescent, ambient air

Methodology: The Numerical Model

The present simulations are performed in the Multi-Component Flow Code (MFC)

- Open-source, high-order, compressible multiphase solver, highly parallel

Resolved Eulerian phase interface:

- Reduced five-equation diffuse-interface formulation of Allaire et al. 2002.

$$\frac{\partial(\alpha_i \rho_i)}{\partial t} + \nabla \cdot (\alpha_i \rho_i \boldsymbol{u}) = 0, \qquad i = 1, 2, \dots N$$

$$\frac{\partial(\rho \boldsymbol{u})}{\partial t} + \nabla \cdot (\rho \boldsymbol{u} \otimes \boldsymbol{u} + p\boldsymbol{I}) = 0$$

$$\frac{\partial(\rho E)}{\partial t} + \nabla \cdot [(\rho E + p)\boldsymbol{u}] = 0$$

$$\frac{\partial \alpha_i}{\partial t} + u \cdot \nabla \alpha_i = K \nabla \cdot \boldsymbol{u}$$

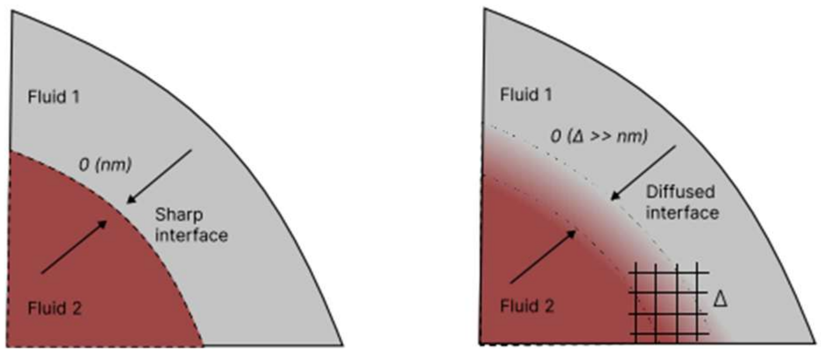


Figure: Sharp and diffuse interface of a multiphase flow (Adebayo 2025).

Methodology: The Numerical Model

Each material is closed with a stiffened-gas equation of state

$$p_i = (\gamma_i - 1)\rho_i e_i - \gamma_i \pi_{\infty,i}$$

The material properties are listed below:

Quantity	Air	Liquid water
Density, ρ	1.17 kg m^{-3}	1000 kg m^{-3}
Ratio of specific heats, γ	1.4	4.4
Stiffened pressure, π_∞	0	600 MPa

Table: Equation-of-state and material parameters used in the shock-droplet simulations.

Viscous stresses and surface tension are neglected in the resolved Eulerian equations.

Methodology: Numerical Methods

The Eulerian equations are discretized on a structured grid using a finite-volume formulation.

Numerical Component	Method Used
Face reconstruction	2^{nd} order MUSCL + van Leer limiter
Interface treatment	THINC interface sharpening
Riemann Solver	HLLC
Time-integration	SSP-RK3 (TVD)

Table 3.2 Summary of numerical methods used for the Eulerian flow solver and sub-grid bubble dynamics.

A constant CFL value of 0.1 is used in all simulations.

Figure: Numerical results for a planar Mach onto a gas bubble. With THINC (left) and without THINC (right) are both shown (Shyue-Xiao 2014).

Methodology: Benchmarking

Benchmarking Against Shock-Water Column Experiments:

- Sembian 2016 studies the interaction of a planar shock (Mach = 1.75 and 2.4) onto a droplet column of diameter $D_0 = 22\ mm$
- Large droplet allows for the observation of key internal acoustics
- Numerical schlieren is compared directly to the experimental shadowgraph
  - Comparison of transmitted shocks, reflected waves, expansion regions, slip surfaces, and triple-point/Mach-stem structures
- Excellent agreement is observed

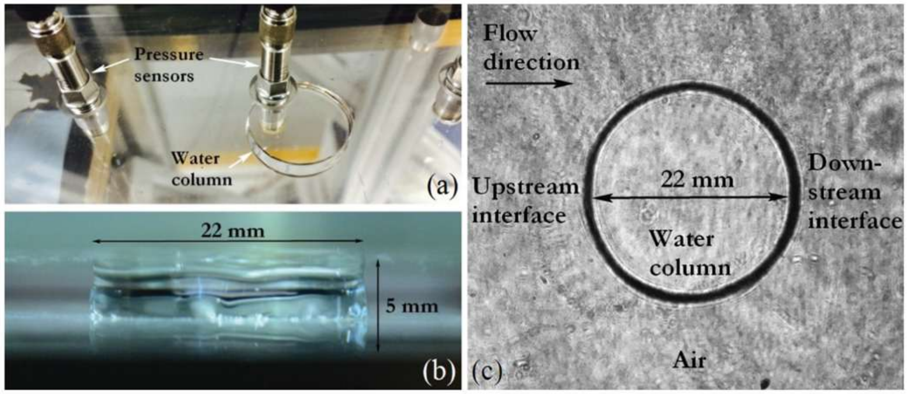


Figure: 22mm water column of Sembian 2016.

Methodology: Validation / Benchmarking

Benchmarking Against Shock-Water Column Experiments:

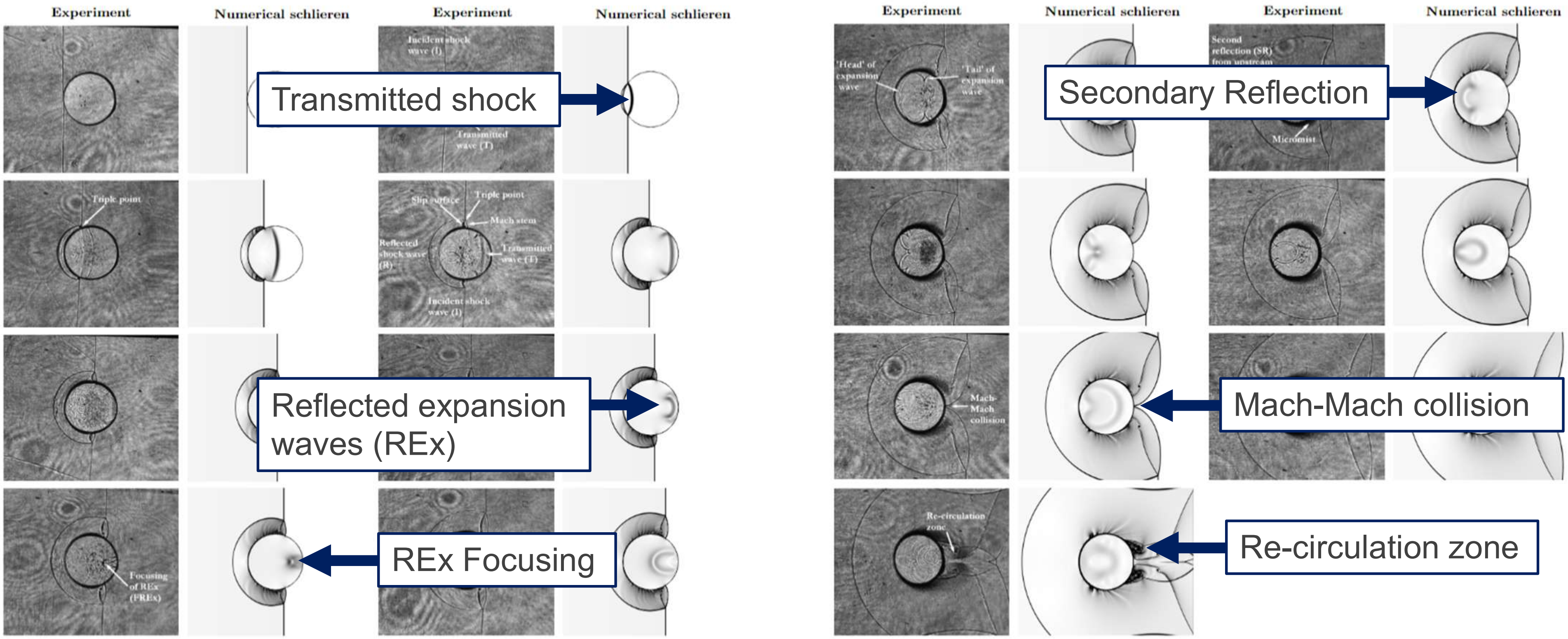


Figure: Comparison between experimental shadowgraph images from the Mach 2.4 shock-water-column experiment of Sembian et al. and numerical schlieren fields from the MFC simulation.

Methodology: Mesh Independence

- Performed on a clean droplet in our case of interest
- Mach 7 is chosen for the study as it produces the largest pressure gradients within the droplet
- A refinement factor of $\sim\sqrt{2}$ is used

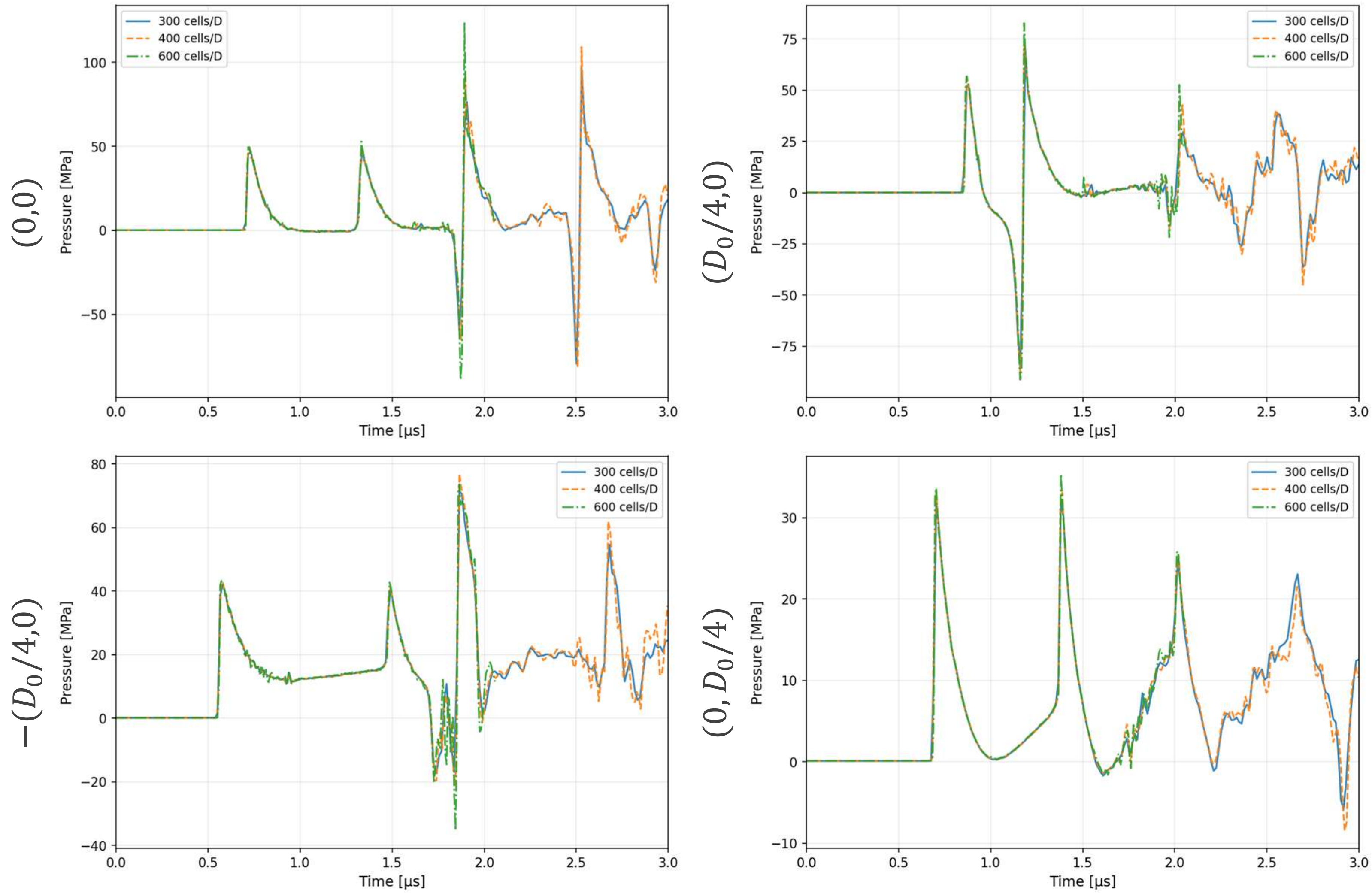
Mesh	Cells / D_0	Grid size $(N_x \times N_r)$	Δx
Coarse	300	6000×1200	3.33 μm
Medium	400	8000×1600	2.50 μm
Fine	600	12000×2400	1.67 μm

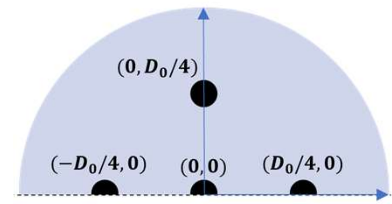
Table: Mesh levels used in the shock-droplet mesh-independence study.

- Shock-arrival time, peak pressure, pressure decay, and area-averaged pressure response across mesh levels was compared
- All meshes show close agreement, hence the medium mesh was chosen

Methodology: Mesh Independence

Nearest-cell pressures





Results: Droplet Morphology

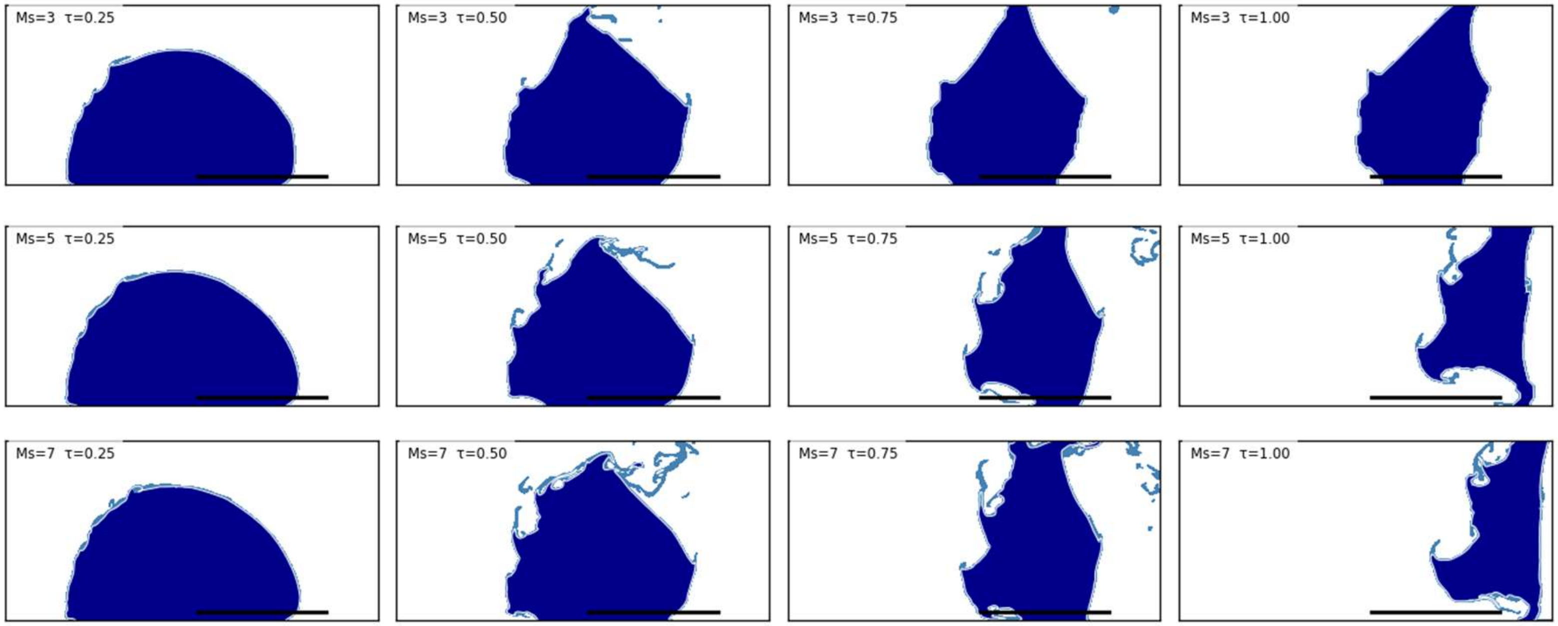
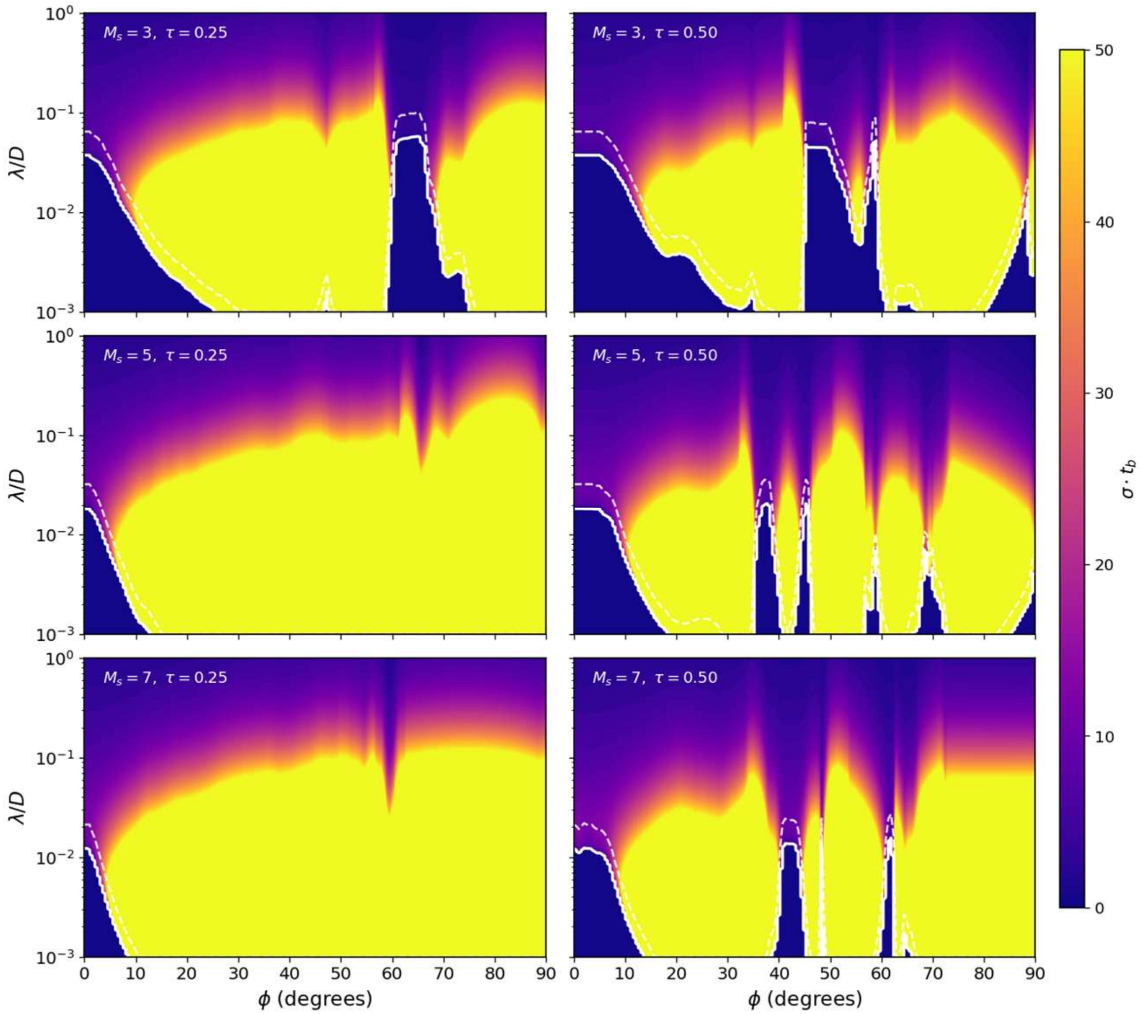


Figure: Liquid volume fraction for $M_s = 3$ (top), 5 (middle), 7 (bottom) at $\tau = 0.25, 0.5, 0.75, and\ 1.00$.

- All quantities are presented in the dimensionless time, t_b is the Ranger-Nichols aerobreakup timescale:

$$\tau = \frac{t - t_{arr}}{t_b}, \qquad t_b = \frac{D}{U}\sqrt{\frac{\rho_l}{\rho_g}}$$

Results: CFD-Informed Growth-Rate Maps



- Growth rate increases monotonically toward smaller λ
- Unstable band broadens with increasing Ms
- RT dominates near stagnation, KH takes over toward the equator
- Narrow re-stabilization bands appear at later times, as the droplet flattens

Figure: FJ growth rate $Re(s) \cdot t_b$ as a function of polar angle and normalized wavelength. White solid: stability boundary Re(s) = 0. White dashed: RT-to-KH transition angle.

Results: RT to KH Transition

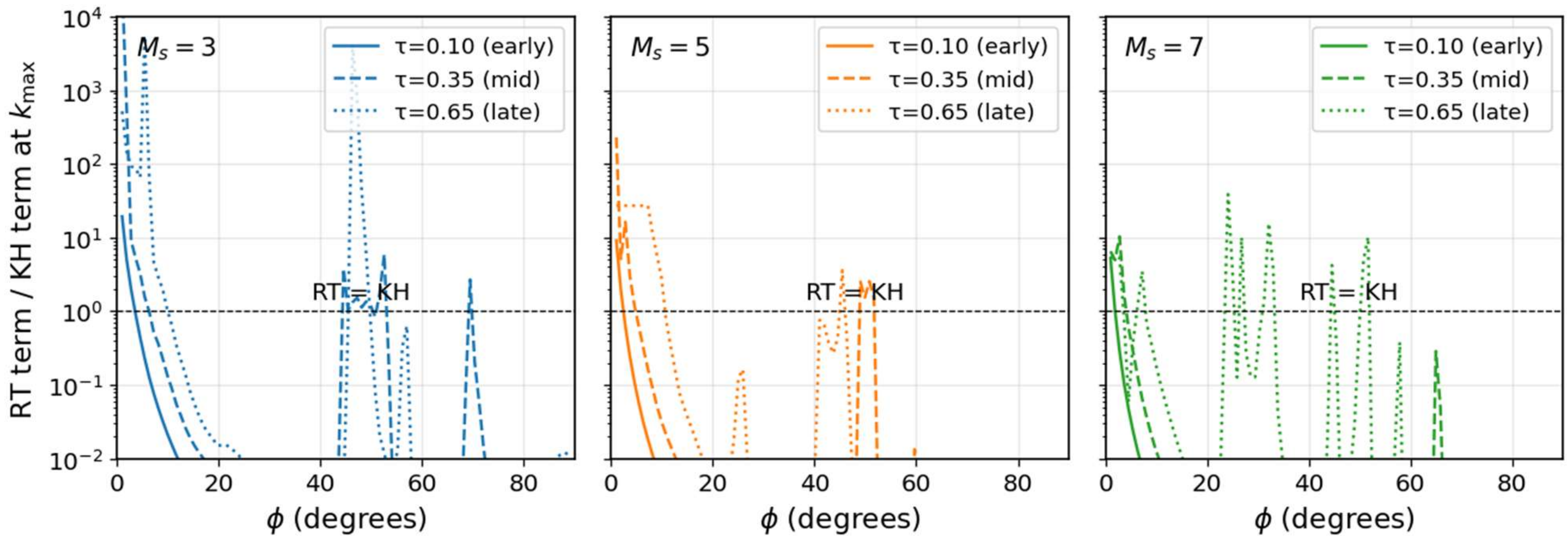


Figure: Ratio of RT acceleration terms to KH shear term evaluated at κ^*.

- At early times, KH shear dominates over most of the front face
- As the droplet flattens over time, RT can dominate over more of the drop surface
- KH appears to dominate at higher Machs (likely wrong)

Results: Most-Amplified Wavelength

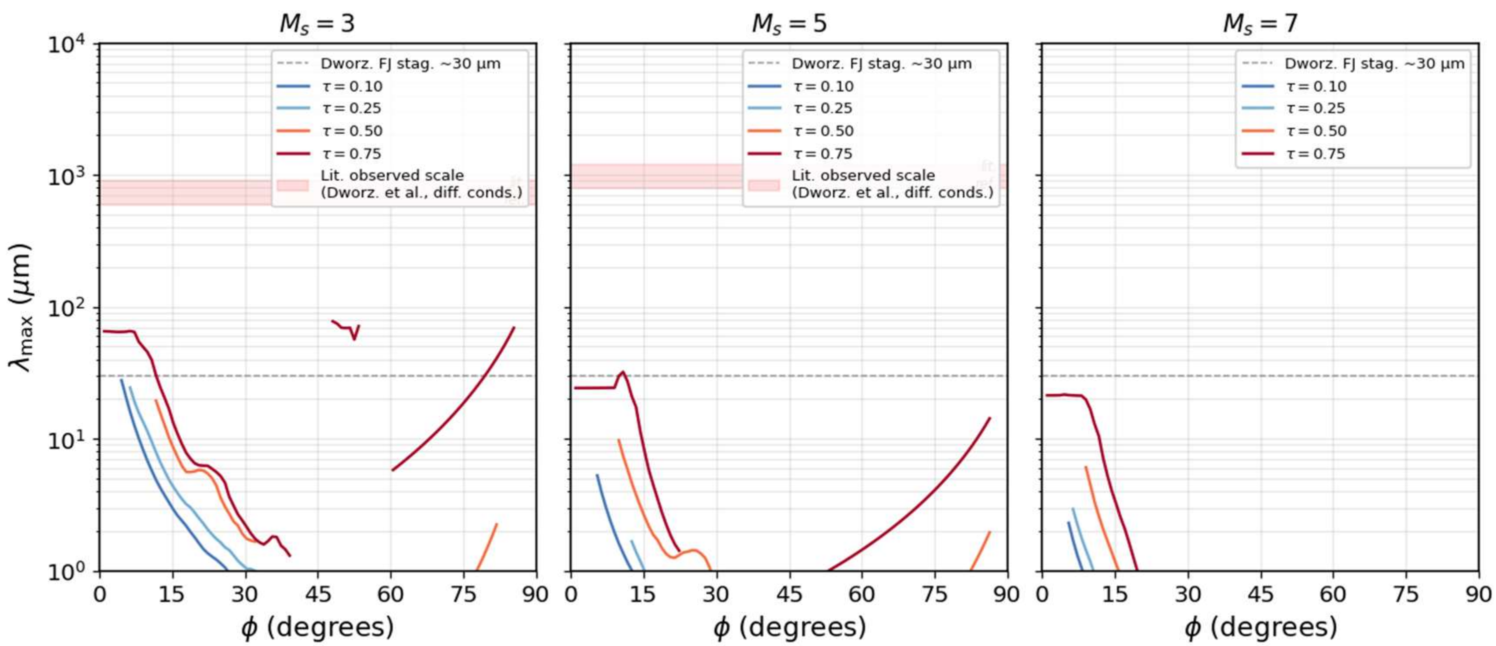


Figure: Most amplified wavelength λ^* *from the FJ dispersion relation.*

- Across all wavelengths, the predicted λ is very small
- Increasing Mach number shifts even smaller
- Linear theory identifies where growth is strong, but it does not predict the observed dominant breakup wavelength

Discussion: Role of Surface Tension

- At large observed structures, surface tension is weak
- At small linear instabilities, surface tension strongly damps high-*k* modes
- Although viscosity and surface tension are negligible at the observed wavelength, they may be needed to create the finite linear peak
- Without surface tension, the simulation has no physical high-*k* cutoff
  - Re(s) increases toward smaller wavelengths
  - No physical most-amplified wavelength can emerge from the CFD
  - Finite surface textures likely tied to grid effects

Summary of Findings

- CFD-informed base states produce useful RT/KH growth maps
- RTI is favored near the windward face; KHI becomes important toward the rim
- Increasing Ms broadens unstable wavelength bands and accelerates deformation
- CFD inputs add spatial/time detail, but do not fix wavelength prediction

Conclusion

- Linear theory predicts $\lambda^{*} \sim 10 - 100\ \mu m$
- Observed structures are much larger: $\sim 500 - 1000\ \mu m$
- Visible wavelength selection is likely finite-time, nonlinear, and initial-condition sensitive

Future Work

- Run 3D simulations with physical surface tension (and viscosity)
- Prescribe controlled single-mode perturbations
- Measure transition time and nonlinear growth rate directly
- Compare to experimental images more closely
- Test mesh effects (resolution, cylindrical vs. Cartesian)
- Sharp vs. diffuse interface